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# Portfolio Optimization Based on Signed Graph Convolutional Network

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## Abstract

In the field of investment, it is crucial to deeply understand the nonlinear relationships among financial tools such as risk management, portfolio construction, and trading strategies. This study focuses on the interdependencies based on stock correlation matrices, constructing them as a network model where individual nodes represent distinct stocks, and weighted links between node pairs reflect their correlation coefficients. Although traditional network science techniques are widely applied in financial research, they often require manual extraction of features (e.g., centrality metrics) to analyze such correlation networks, a process that is complex and cumbersome. Therefore, this study employs a Signed Graph Convolutional Network to explore signed connections within correlation networks. This algorithm compresses the network into a low-dimensional, continuous embedding space, where similar nodes are positioned closer to one another. Using this approach, we screen stocks and apply the selected stocks to portfolio optimization. Through the return and risk measures of the resulting portfolio, this study demonstrates that the proposed algorithm can learn effective embedding representations from the correlation network. Furthermore, the research explores the diverse potential applications of embeddings in the field of investment management.

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**Keywords:** stocks; machine learning; graph embedding; clustering; unsupervised learning

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## 1. Introduction

Drawing on classical Markowitz theory [1, 2], traditional portfolio optimization revolves around estimating expected returns and covariance matrices, followed by constrained optimization to settle on asset allocations. However,

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later research has uncovered several limitations. For example, Siegel and Woodgate [3] found that mean-variance optimized portfolios often underperform out-of-sample. Guidolin and Ria [4] revealed that unrealistic assumptions, like return normality, zero transaction costs, and constant market cycles, undermine these models in practice. Ao et al. [5] further pointed out that expected returns and covariance matrices, being unobservable, must be substituted with sample estimates, while DeMiguel et al. [6] demonstrated that the traditional models struggle with the high-dimensional nature of modern portfolios.

To navigate the challenges posed by non-convexity and nonlinearity, researchers have turned to deep learning. For instance, Lv et al. [7] introduced a hybrid system that couples stock selection with portfolio optimization using multiple objectives; Kim et al. [8] integrated GANs for anomaly detection within the mean-variance framework; Avramov et al. [9] applied neural networks and conditional auto-encoders to pinpoint mispriced stocks; and Korangi et al. [10] showed that deep learning models can handle multi-objective constraints effectively. Together, these approaches highlight deep learning's promise, especially for mid-cap portfolio optimization.

Advancements in AI have further revolutionized relational representation learning, enabling the detection of deep temporal correlations. Inspired by NLP embedding techniques like Word2Vec [11], which is based on Skip-gram architectures, financial models such as the modified Stock2Vec [12] now jointly embed corporate financial texts and price series. Moreover, Qu et al. [13] crafted a portfolio optimization strategy based on quantum walks, while Shi et al. [14] proposed a portfolio management framework that combines Graph Convolutional Networks with reinforcement learning.

Drawing from these studies, we conclude that deep reinforcement learning clearly outperforms conventional methods in portfolio optimization due to its scalability and ability to transfer learning. At the same time, traditional correlation measures tend to focus solely on linear relationships, thereby overlooking nonlinear and negative interactions, while graph-based methods tend to emphasize positive correlations. Moreover, common sparsification techniques such as the minimum spanning tree suffer from a lack of clear mathematical interpretability.

In response to these challenges, we propose a portfolio optimization method that integrates Spearman coefficients, signed graph neural networks, adaptive LASSO sparsification, and reinforcement learning.

## 2. Theoretical Foundations

Markowitz's Modern Portfolio Theory (MPT) established the risk-return trade off framework in modern finance, [1,2]. Formally, the expected return of a portfolio is expressed as a linear combination of individual asset returns:

$$E(R_p) = \mathbf{w}^T \boldsymbol{\mu} = \sum_{i=1}^n w_i \mu_i \quad (2.1)$$

Where  $E(R_p)$  represents the expected return of the portfolio,  $\mathbf{w}$  is the weight vector satisfying  $\sum_{i=1}^n w_i = 1$ , and  $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_{1n})^T$ , is the vector of expected returns for individual assets. Portfolio volatility, measured as the variance of returns, is represented by the quadratic form of the variance-covariance matrix:

$$\sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w} = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} \quad (2.2)$$

Where  $\Sigma$  represents the correlation matrix of the assets,  $\sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$  denotes the covariance between assets  $i$  and  $j$ , and  $\rho_{ij}$  is their correlation coefficient. The optimization objective is to minimize risk or maximize returns, mathematically formulated as:

$$\begin{cases} \min \mathbf{w}^T \Sigma \mathbf{w} \\ \text{s. t. } \mathbf{w}^T \boldsymbol{\mu} = E(R_p) \\ \mathbf{w}^T \mathbf{1} = 1 \end{cases} \quad (2.3)$$

### 3. Methodology

As shown in Figure 1, we used quarterly data from 2020 – 2025 to compute risk measures under computational constraints. We derived an association matrix via Spearman rank correlation on quarterly stock prices, obtained embedded representations using a signed graph convolutional network, merged these with risk measures, screened the data using clustering, and finally optimized the portfolio with deep reinforcement learning.

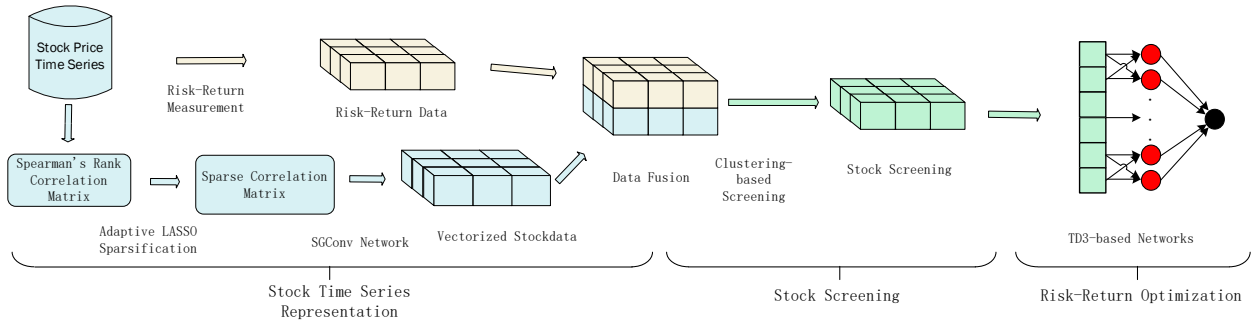


Fig. 1. Method Schematic Diagram.

#### 3.1 Sparse Correlation Matrix Based on Adaptive Lasso

##### 3.1.1 Spearman Correlation Coefficient

Empirical evidence shows significant nonlinear stock dependencies, as confirmed by our dataset [15]. Therefore, we use Spearman's rank correlation coefficient [16] to quantify stock price associations. The coefficient is defined as:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (3.1)$$

where  $d_i$  is the difference in ranks between paired observations, and  $n$  is the sample size.

##### 3.1.2 Adaptive Lasso Estimation

The LASSO method employs L1 penalization within the least squares framework for coefficient shrinkage, as introduced by Tibshirani [17]. However, its uniform penalization of all regression coefficients is a key limitation. To overcome this, Huang et al. [18] developed the adaptive LASSO, which assigns coefficient-specific weights to the penalty term. The formulation is:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - \sum_{j=1}^p \mathbf{x}_i^T \beta_{ij})^2 + \lambda \sum_{j=1}^p w_j |\beta_j| \right\} \quad (3.2)$$

where  $\lambda$  is the regularization parameter,  $n$  is the sample size, and  $p$  is the number of variables.  $\mathbf{x}_j$  represents the rank-transformed price series of the  $j$ -th stock (derived from the Spearman correlation matrix),  $y_i$  represents the asset corresponding to the  $i$ -th row, and  $w_{ij}$  is an adaptive weight designed to impose differential penalization. and  $w_{ij}$  is an adaptive weight that imposes differential penalization: smaller penalties for predictors with stronger associations and larger ones for less relevant variables. We innovatively define the adaptive weights as follows:

$$w_{ij} = \frac{1}{|\hat{\beta}_{ij}^{\text{init}}| + \varepsilon} \quad (3.3)$$

Where  $\hat{\beta}_{ij}^{\text{init}}$  denotes the Spearman correlation coefficient between stocks  $i$  and  $j$ , modified by setting all diagonal elements (self-correlations) to zero to eliminate self-interference. The term  $\varepsilon$  is a small constant ( $\varepsilon = 10^{-4}$ ) added for numerical stability. This weighting scheme ensures that higher correlations ( $|\hat{\beta}_{ij}^{\text{init}}|$ ) correspond to weaker penalties. Additionally, to eliminate interference caused by the main diagonal (i.e., self-correlation of assets), we set all off-diagonal elements to zero in the matrix ( $\hat{\beta}_{ij}^{\text{init}} = 0$ ). In this way, we preserve the critical relationships among assets while ensuring that the sparsification process possesses strong interpretability and mathematical significance.

### 3.2 Vectorizing the Matrix Using SGConv

SGConv stands for the Signed Graph Convolutional Network. In our study, we constructed a two-layer network to learn the relational structure. The experimental model comprises two layers, with the first propagation layer containing 64 parameters, represented as:

$$\mathbf{H}^{(1)} = \tanh(\mathbf{A}_s \mathbf{H}^{(0)} \mathbf{W}^{(1)}) \quad (3.4)$$

where  $\mathbf{H}^{(0)} = \mathbf{1} \in \mathbb{R}^{N \times 1}$  (initialized with unit vectors) and  $\mathbf{A}_s[i, j] = w_{ij}$  is the signed adjacency matrix,  $\mathbf{W}^{(1)}$  is a learnable parameter matrix of the first layer.

The second layer is an aggregation layer with a total of 32 dimensions and can be expressed as:

$$\mathbf{H}^{(2)} = \mathbf{A}_b \mathbf{H}^{(1)} \mathbf{W}^{(2)}$$

where  $\mathbf{A}_b[i, j] = 1 (w_{ij} \neq 0)$  denotes the binarized adjacency matrix and  $\mathbf{W}^{(2)} \in \mathbb{R}^{d_h \times d_o}$  ( $d_o = 32$ ) represents the associated learnable parameters. The unsupervised loss function employed in this experiment is given by:

$$\mathcal{L} = - \underbrace{\frac{1}{|E^+|} \sum_{(i,j) \in E^+} \cos(\mathbf{h}_i, \mathbf{h}_j)}_{\text{Positive edge constraints}} + \underbrace{\frac{1}{|E^-|} \sum_{(i,j) \in E^-} \cos(\mathbf{h}_i, \mathbf{h}_j)}_{\text{Negative edge constraints}} \quad (3.5)$$

In the loss function of the signed convolutional network,  $E^+, E^-$  represents the set of positive edges and negative edges. On the other hand,  $\mathbf{h}_i, \mathbf{h}_j$  denotes the learned feature vectors or embeddings for the nodes.

### 3.3 CVaR Risk Measurement

CVaR (Conditional Value-at-Risk) is a risk measure introduced by Rockafeller and Uryasev [20] and is considered superior to VaR. It represents the average loss beyond VaR, which in this experiment is computed at a 95% confidence level. VaR indicates the maximum potential loss at that confidence level:

$$\text{VaR}_\alpha = \inf\{z \mid \mathbb{P}(X \leq z) \geq \alpha\} \quad (3.6)$$

Let the random variable  $X$  denote the portfolio loss. VaR then represents the maximum loss at the confidence level  $\alpha$ . Its mathematical definition is given by:

$$\text{CVaR}_\alpha = \mathbb{E}[X \mid X \geq \text{VaR}_\alpha] = \frac{1}{1 - \alpha} \int_{\text{VaR}_\alpha}^{\infty} x p(x) dx \quad (3.7)$$

where  $p(x)$  is the probability density function of the loss.

Boudt et al. [21] proposed using the contributions of individual assets to the portfolio CVaR as inputs to the portfolio optimization problem. The optimization objective for the CVaR contribution percentage of asset  $i$  can be

derived from the weight vector  $\mathbf{w}$ , the mean vector  $\boldsymbol{\mu}$ , and the covariance matrix  $\boldsymbol{\Sigma}$ . The corresponding optimization objective is given by:

$$\min \frac{w_i[-\mu_i + \frac{(\sum \mathbf{w})_i \phi(Z_\alpha)}{\sqrt{\mathbf{w}' \boldsymbol{\Sigma} \mathbf{w} \alpha}]}]{-\mathbf{w}' \boldsymbol{\mu} + \sqrt{\mathbf{w}' \boldsymbol{\Sigma} \mathbf{w} \frac{\phi(Z_\alpha)}{\alpha}}}}{\alpha}, \text{ subject to } \sum w_i = 1, 0 \leq w_i \leq 1 \quad (3.8)$$

where  $Z_\alpha$  is the quantile of the standard normal distribution at the confidence level, and  $\phi(\cdot)$  is the standard normal density function.

### 3.4 Portfolio Optimization

We employed a TD3-based[22] deep reinforcement learning network. Additionally, random noise is added to actions to boost exploration for tackling complex optimization tasks.

#### 1) State Space

The model's input features include the current asset weights  $\mathbf{w}$ , the covariance matrix  $\boldsymbol{\Sigma}$ , historical asset returns  $\boldsymbol{\mu}$ , and risk measure **CVaR**.

#### 2) Action Space

The output action is the adjustment amount of the asset weights  $w_i$ , which must satisfy  $\sum_{i=1}^N \Delta w_i = 0$ , the new asset weight vector fulfills  $w_i \in [0,1]$ .

#### 3) Reward Function

The current implementation of the deep reinforcement learning model allows for constraints in the domain space; we incorporate these constraints into the reward function via penalty functions. A standard expression for these penalties is as follows:

$$\lambda \cdot |\text{violation}|^p \quad (3.9)$$

Here, a penalty function violation =  $|\sum w_i - 1|$  is applied to violations of  $\sum w_i = 1$  or  $w_i \in [0,1]$ .

Research by Crama and Schyns [23] describes several calibration factors,  $\lambda$  and  $p$ . In our experiment we set the parameter values to  $\lambda = 1000$  and  $p = 2$ . The final reward function is thus obtained as:

$$R = -\frac{w_i[-\mu_i + \frac{(\sum \mathbf{w})_i \phi(Z_\alpha)}{\sqrt{\mathbf{w}' \boldsymbol{\Sigma} \mathbf{w} \alpha}]}]{-\mathbf{w}' \boldsymbol{\mu} + \sqrt{\mathbf{w}' \boldsymbol{\Sigma} \mathbf{w} \frac{\phi(Z_\alpha)}{\alpha}}}}{\alpha} - \lambda \cdot |\sum w_i - 1|^2 \quad (3.10)$$

## 4. Empirical Research

In this study, we use publicly available CSI300 stock return data from the jqdata platform. After cleaning the data and constructing a stock network, we apply k-means clustering to the stock embedding vectors, selecting the optimal number of clusters via the silhouette coefficient. From each cluster, we then rank stocks by the Sharpe ratio and choose the top 50% according to three segmentation strategies: the most central 50%, a combination of the most central 25% and the most peripheral 25%, and the most peripheral 50%. These selected stocks form the portfolio.

#### 4.1. Analysis of Stock Embedding Results

The Tonghuashun stock industry classification is used as a reference, grouping stocks hierarchically with each CSI 300 stock receiving a unique classification value.

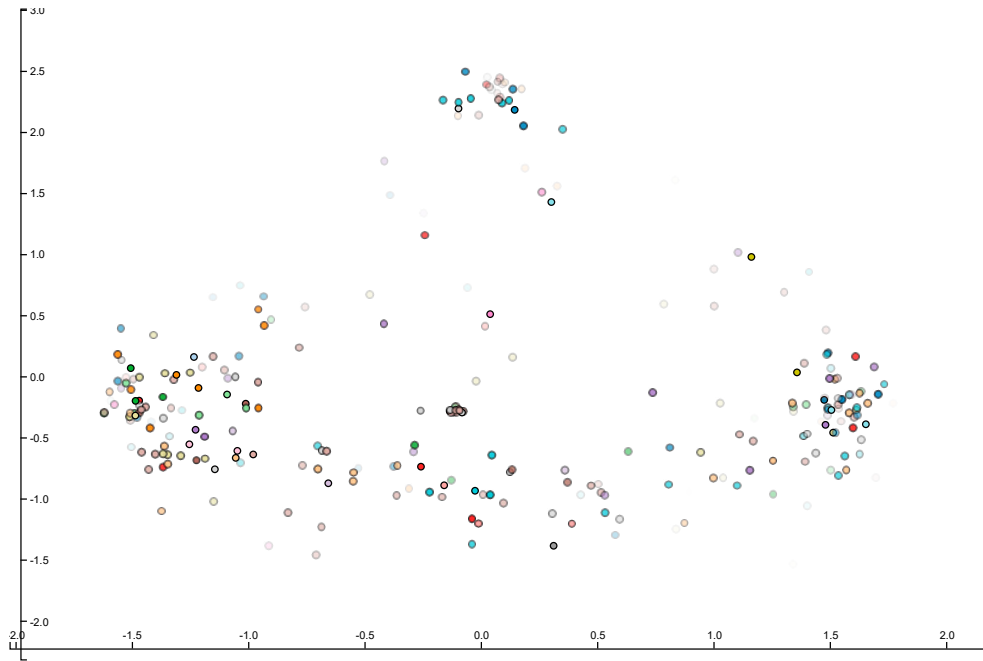


Fig. 2. PCA Dimensionality Reduction Results.

Our study reveals that stock embeddings are more sensitive to macroeconomic changes than static industry classifications. For example, in Q2 2023, while Pacific Life Insurance's embedding indicates a financial sector correlation, its classification does not strictly align with this trend. PCA results further show that stocks with similar Tonghuashun sector classifications form distinct clusters. To demonstrate our approach's advantages, we constructed a portfolio by selecting the top 25% Sharpe ratio stocks within each sector, compared to selecting the top 25% from the entire group.

#### 4.2. Baseline Methods

To validate our GCN-based approach, we used spectral clustering [24] as the baseline. We computed the first  $k$  eigenvectors to form matrix  $U$ , normalized its rows, and then applied the same elbow method-based K-means clustering on the normalized  $U$  before training the model.

#### 4.3. SGCN-Based Sampling Method

Although the Tonghuashun labels did not boost model performance, we used the V-measure from Tonghuashun classification for hyperparameter tuning, employing the Adam optimizer with a 0.01 learning rate and a weight decay of  $1e-4$ . To validate the Spearman coefficient, we conducted an identical experiment with the Pearson correlation. Despite using the same adaptive Lasso sparsification, the average sparsity was 61.58% for Spearman versus 88.34% for Pearson. Our results are as follows:

Table 1: Display of All Results

| Sampling Method                                      | Annualized Return | Volatility | Maximum Drawdown |
|------------------------------------------------------|-------------------|------------|------------------|
| CSI300 Index                                         | 1.51%             | 18.85%     | -17.33%          |
| Random Sampling                                      | 1.93%             | 21.02%     | -44.63%          |
| Tonghuashun Sector Classification Sampling           | 2.16%             | 24.02%     | -51.18%          |
| Spectral Clustering - Up & Down Sampling             | 5.36%             | 17.85%     | -34.52%          |
| Spectral Clustering - Up Sampling                    | 2.56%             | 19.25%     | -45.63%          |
| Spectral Clustering - Downsampling                   | 2.89%             | 20.53%     | -43.76%          |
| Experimental Method - Up & Down Sampling             | 6.54%             | 18.37%     | -38.02%          |
| Experimental Method - Upsampling                     | 2.73%             | 22.42%     | -41.37%          |
| Experimental Method - Downsampling                   | 2.89%             | 20.53%     | -43.76%          |
| Pearson Correlation Coefficient - Up & Down Sampling | 5.84%             | 20.42%     | -40.29%          |
| Pearson Correlation Coefficient - Upsampling         | 3.23%             | 20.71%     | -38.67%          |
| Pearson Correlation Coefficient - Downsampling       | 2.42%             | 18.53%     | -32.12%          |

It can be seen that SGCN outperforms spectral clustering in terms of annualized returns, although its volatility and maximum drawdown are slightly higher. Spectral clustering, based on graph theory, relies on the eigen-decomposition of the Laplacian matrix to cluster stocks, making it suitable for capturing global structural relationships in the market. However, it is less efficient when processing high-dimensional sparse data and often struggles to capture intricate relationships. In contrast, SGCN dynamically aggregates neighborhood information of nodes (stocks) through graph convolution operations, enabling it to capture local nonlinear relationships and complex market dynamics. Through end-to-end training, SGCN automatically learns the latent associations between stocks without the need to manually define a similarity matrix, thereby offering better adaptability to noisy data and temporal changes. Moreover, SGCN can be integrated into a deep reinforcement learning framework to directly incorporate risk control (such as maximum drawdown constraints) and transaction costs into the decision-making process, achieving a more refined risk-return balance.

Experimental results show that, compared to methods based on Pearson's correlation coefficient, which measure only linear relationships, the rank-based Spearman correlation metric captures richer information, ultimately achieving better performance in portfolio optimization. Thanks to the fact that adaptive LASSO directly computes and penalizes correlations, our strategy, compared to methods such as the Minimum Spanning Tree and Planar Maximally Filtered Graph, can retain important relationships while maintaining good interpretability.

Overall, SGCN possesses stronger capabilities in modeling nonlinear relationships, dynamic risk adjustment, and end-to-end optimization, which allow it to perform better in complex market environments. Furthermore, this study confirms that the inclusion of edge or weakly correlated assets indeed improves the performance of portfolio optimization.

## 5. Conclusion

In summary, our study shows that a signed graph convolutional network-based representation learning approach not only captures stock associations effectively with clear interpretability but also provides enhanced correlation measurements over traditional methods. Future research should explore the benefits of dynamic stock selection and incorporate dynamic, multimodal optimization, particularly in high-frequency trading scenarios that account for transaction fees, liquidity risks, and state-dependent adjustments. Although our experiments focused on CSI 300 stocks, our approach is easily extendable to international markets.

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